Experiment-5 Report

AVR IoT

R. Vikas Raj, EE19B108, Batch-1 October 2024

Objectives:

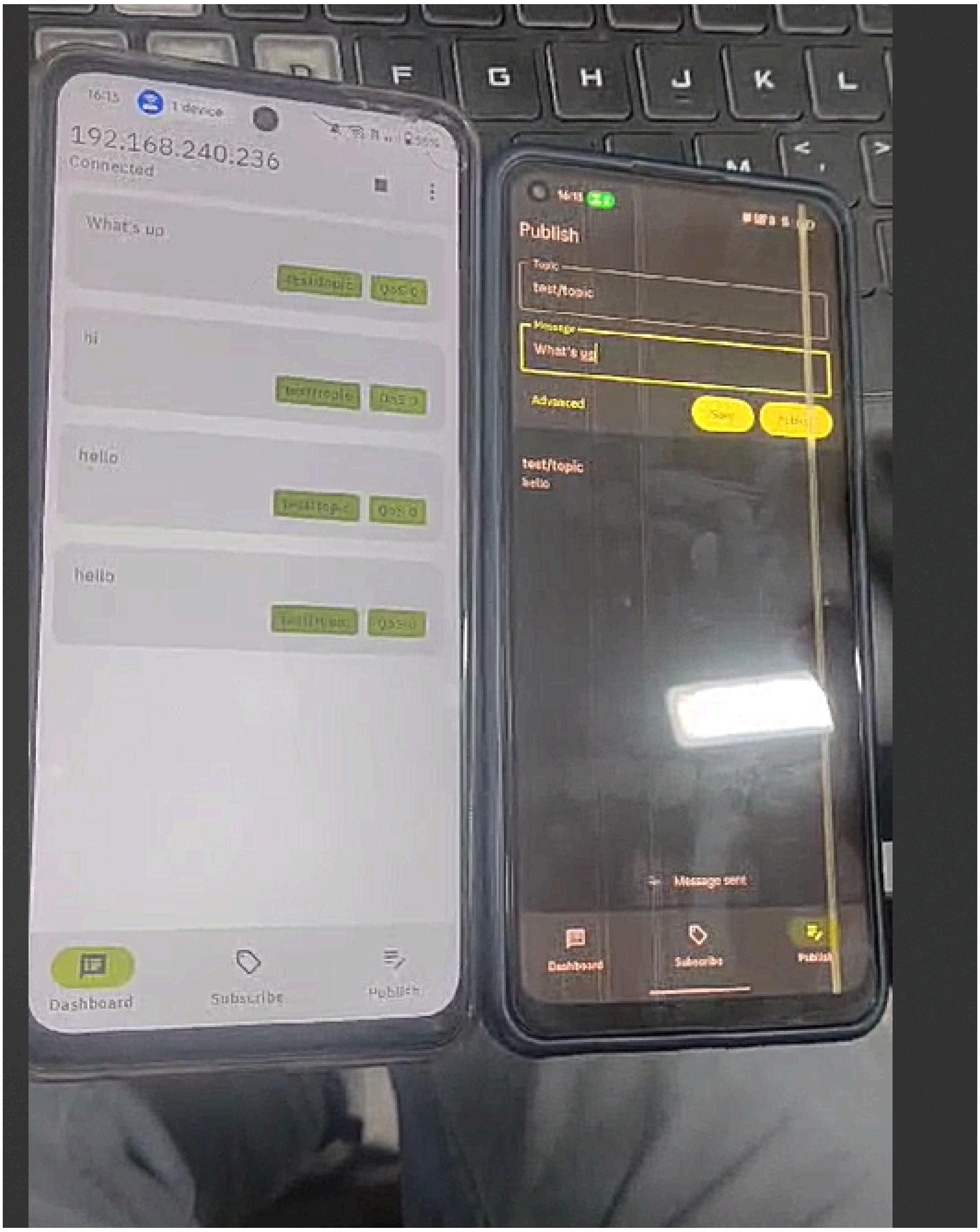
- To implement two IoT applications using an AVR-based microcontroller and MQTT protocol, one for board-to-board communication and the other for text publishing.
- To understand publishers and subscribers using the Mosquitto server and My MQTT app.

Hardware and Software Used:

- AVR IoT boards (2)
- USB-B cables (2)
- LCD screen (1)
- Jumper wires
- Mosquitto broker
- MPLAB X IDE

Machine-to-Machine (M2M) communication:

using the MQTT protocol, this sub-experiment involved setting up a Mosquitto Broker on a laptop and connecting two mobile devices to it using the MyMQTT app. All devices were connected to the same Wi-Fi network, with the broker address set to the laptop's IPv4 address. One mobile device published a message to a specific topic (e.g., "test/topic"), and the second device, subscribed to the same topic, received the message in real-time. The Mosquitto broker facilitated this communication by logging client actions such as connections, subscriptions, and message publishing. This experiment demonstrated the functionality of the MQTT protocol for efficient, real-time data transmission, highlighting its potential for broader IoT applications where multiple devices need to communicate seamlessly.



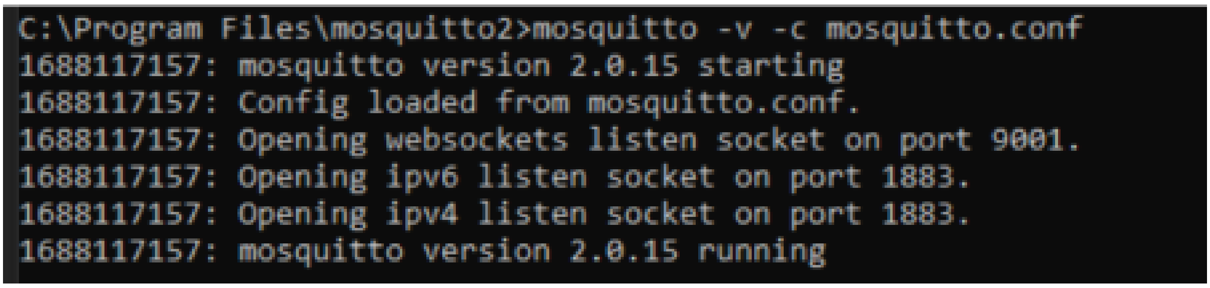
MQTT Protocol Overview

Message Queuing Telemetry Transport (MQTT) is a communication protocol frequently used in IoT applications. It is preferred over AMQT and UDP due to its efficient publisher-subscriber architecture. The protocol enables communication between devices by transmitting data through topics, with a central broker managing TCP connections. Devices can subscribe to or publish information on topics, enabling seamless data sharing within the network.

Setting Up Mosquitto Broker

Mosquitto is used to set up the PC as a broker for MQTT. Configuration is done by modifying the `mosquitto.conf` file with the following parameters:

```
listener 9001
protocol websockets
listener 1883
allow anonymous true
password file C:\Program Files\mosquitto\pwfile.example
```



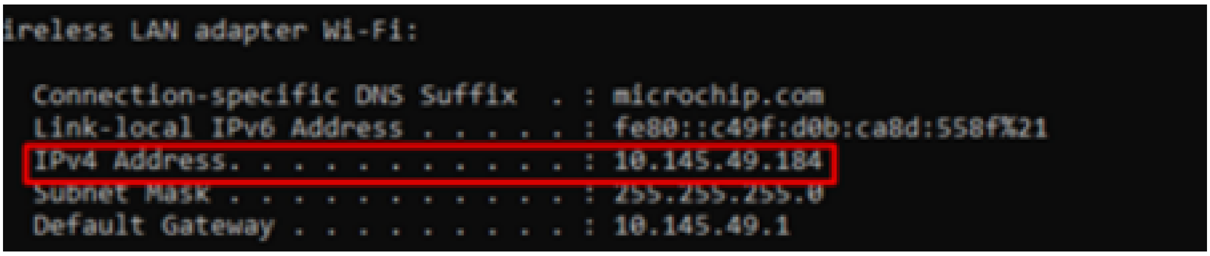
```
C:\Program Files\mosquitto2>mosquitto -v -c mosquitto.conf
1688117157: mosquitto version 2.0.15 starting
1688117157: Config loaded from mosquitto.conf.
1688117157: Opening websockets listen socket on port 9001.
1688117157: Opening ipv6 listen socket on port 1883.
1688117157: Opening ipv4 listen socket on port 1883.
1688117157: mosquitto version 2.0.15 running
```

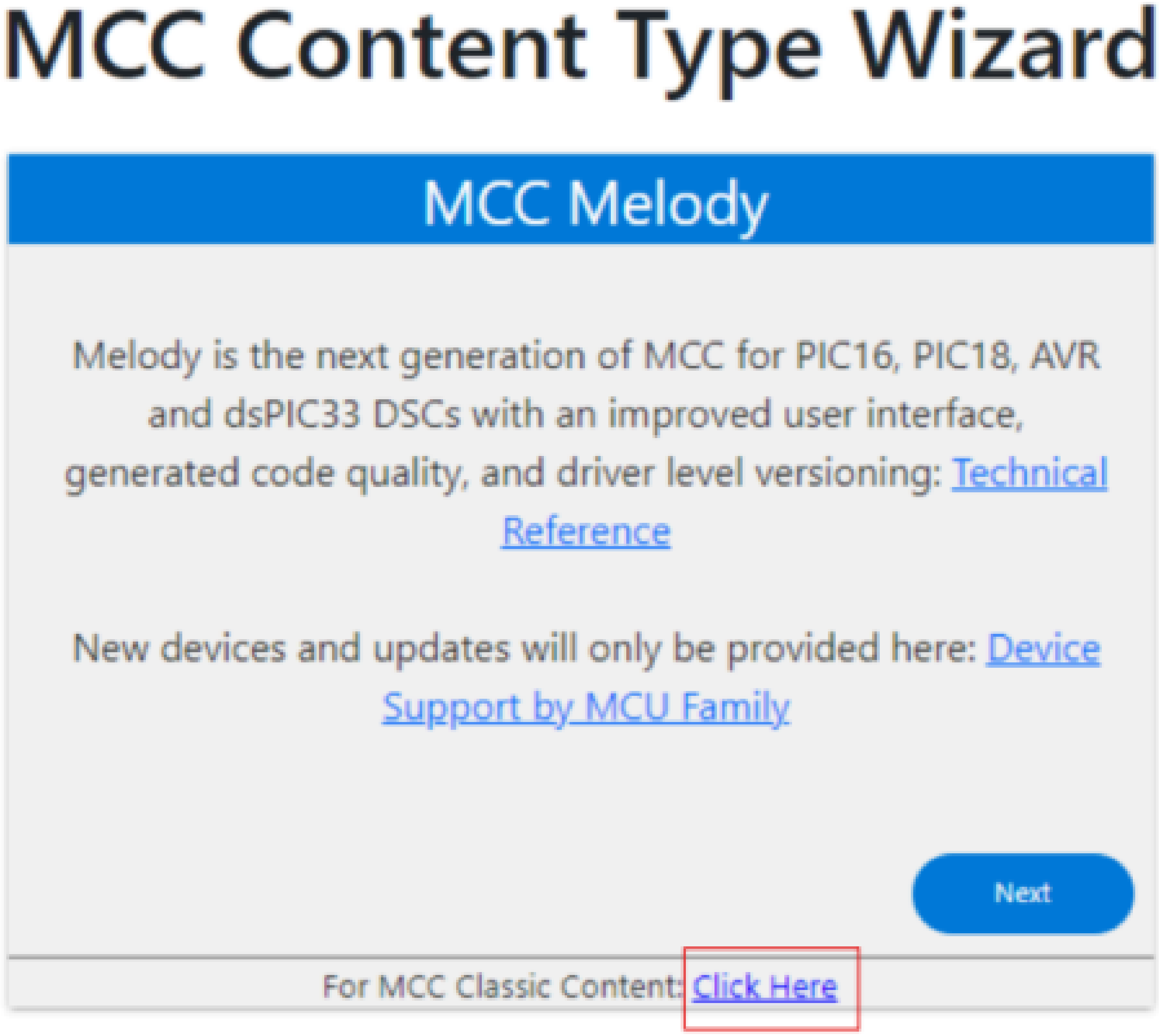
To run the broker, use the command:

```
mosquitto -v -c mosquitto.conf
```

AVR-IoT Board Details

The AVR-IoT board, powered by an Atmega4808 processor, was used for this experiment. It includes a built-in Wi-Fi module for MQTT communication and several GPIO pins for connecting additional devices, such as the LCD screen.



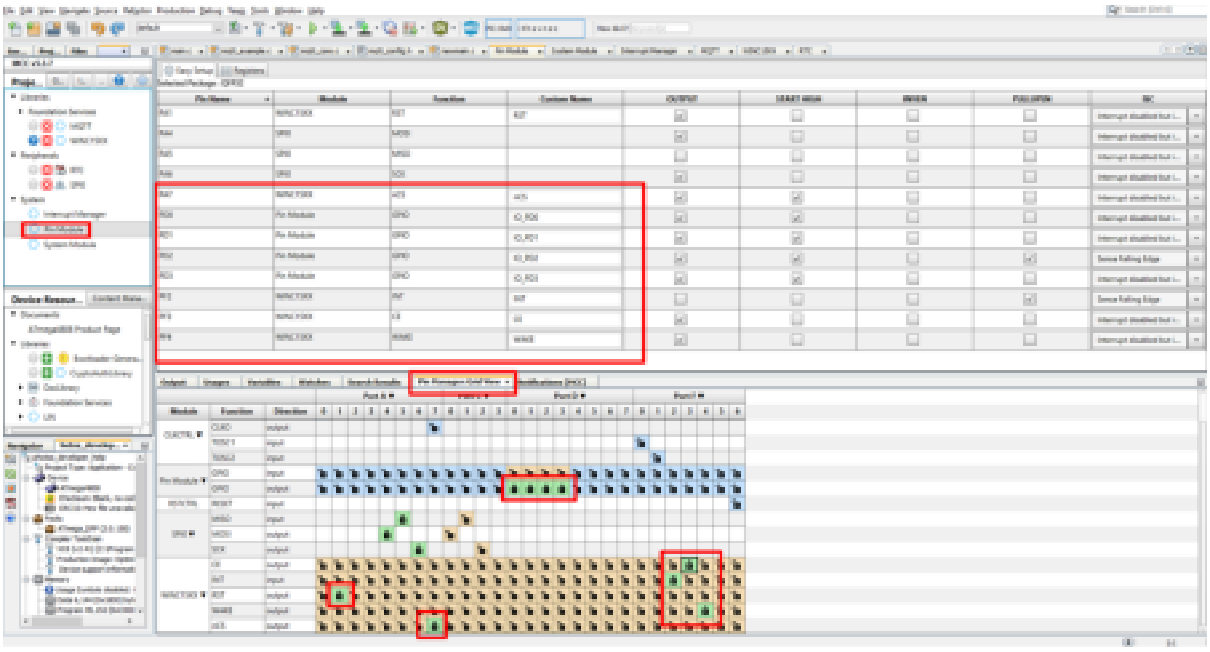


Mobile Demonstration of MQTT

To demonstrate MQTT using mobile devices:

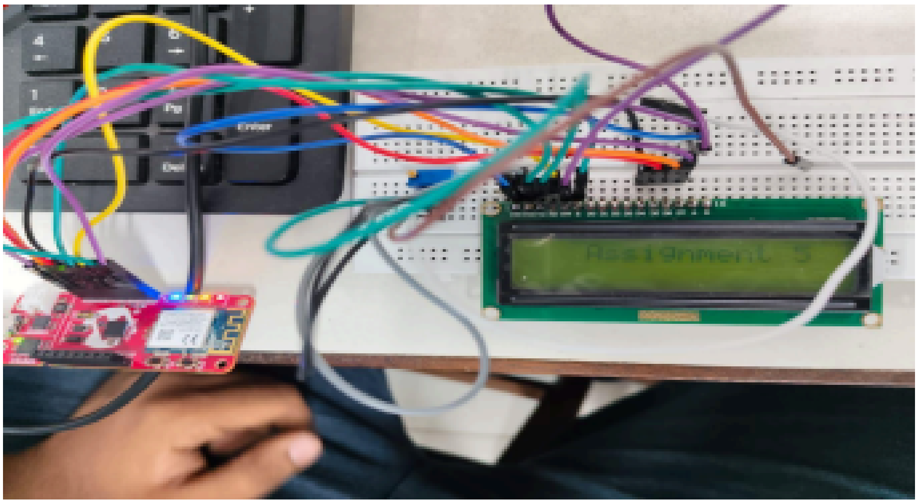
1. Install the "My MQTT" app on two phones.
2. Set up Mosquitto as a broker.
3. Connect both mobiles to the same network as the broker. 4. Enter

the broker’s IP address found via `ipconfig` and connect. 5. Create a topic, subscribe both phones, and publish a message. The second device will receive the message.



AVR-IoT Board Implementation

1. Install the MQTT library as per instructions.
2. Create a new project in MPLAB X IDE, selecting the Atmega4808 board and the XC8 compiler.
3. Add the MQTT and WINC15XX modules to the project resources and configure them.
4. Build and upload the publish code to Board A and the subscribe code to Board B. Upon success, the LED on Board B will blink.
5. Connect the LCD to the AVR-IoT board, and with further steps, display the message "Assignment 5" on the screen.



Conclusion:

This experiment demonstrated the versatility of AVR boards for IoT applications, emphasizing machine-to-machine communication and interfacing with LCD displays. The MQTT protocol proved effective due to its subscriber-publisher architecture, offering advantages over traditional protocols like UDP.